Brian Wonjun Lee

Philadelphia, PA | 215-294-7152 | leebwj@sas.upenn.edu | leebrian.dev | LinkedIn

Education

University of Pennsylvania — Philadelphia, PA

MSE in Computer Graphics & Game Technology (CGGT), Submatriculation — *Expected May 2028*

BA in Computer Science & Design (Double Major), Benjamin Franklin Scholars — *Expected May 2027*

Relevant Coursework: Interactive Computer Graphics, Computer Animation, 3-D Computer Modeling, Introduction to Algorithms, Introduction to Computer Systems, Programming Languages & Techniques

Experience

Software Engineer Intern, Aleph Lab (Y Combinator) — *June 2026 – Aug 2026 · San Francisco (Remote)*

- Delivered an app-wide **React Native** redesign from Figma to production, sequencing information architecture ahead of the visual system so users kept their bearings, then carrying a **33-section** design system onto every screen
- **Built the company's lifecycle notification system end to end, then made it autonomous** — per-stage copy, anti-nag back-off and a measurement holdout; it clears its own guardrails and delivers unattended at **91% device reach**, and widening eligibility across parent and child accounts reached **~2.7×** more families and its first paid conversion
- **Packaged the in-game AI agent as a versioned SDK outside studios can build on** — typed APIs, per-session isolation, capability enforcement and **1,301 tests**, published over a hosted multi-tenant service — proven by an external builder shipping a mode with no repository access, loading their own content at runtime
- **Closed two child-safety gaps** — public endpoints that trusted any user ID, bearer-authed with **zero caller changes**; and the AI agent's defensive targeting, which had been starting fights with neutrals and bosses far above a child's level

Software Engineer Intern, BitMango (Puzzle1 Studio) — *June 2025 – Aug 2025 · Pangyo, South Korea*

- Maintained a live portfolio of **50+ Unity mobile game titles**, improving UI/UX and core gameplay stability across releases
- Resolved **100+ QA-reported bugs** by modifying Unity prefabs and C# scripts, shipping fixes in production releases
- Upgraded SDKs and platform modules fleet-wide to meet app-store requirements and sustain availability

Project Lead & Product Designer, Penn Spark — *Jan 2025 – Present · Philadelphia, PA*

- Led a cross-functional team of **15+** designers and developers to ship client-facing **0-to-1 products**
- Defined end-to-end user flows, interaction patterns, and reusable interface systems across multiple deliverables
- Shipped polished, user-centered features across semester-long client project cycles

Military Interpreter, Capital Defense Command (ROK Army) — *Dec 2022 – June 2024 · Seoul, South Korea*

- Provided real-time English–Korean interpretation in high-security, multi-agency environments for international stakeholders
- Translated sensitive documents and briefings with precision, enabling coordination across government and defense teams

Database Engineer Intern, IT-Farm — *June 2022 – Aug 2022 · Seongnam, South Korea*

- Designed and implemented a SQL database prototype managing **1M+ data entities** using Oracle and SQLite
- Processed and organized semiconductor datasets from **20+ client companies**, including Samsung, SK, and LG
- Wrote and optimized SQL queries to improve retrieval and management efficiency across large-scale datasets

Projects

Mini Minecraft | *C++, OpenGL, GLSL* — *Apr 2026*

- **Made a from-scratch voxel world read as varied rather than noisy** — **7-biome procedural generation** (Perlin/FBM/Voronoi, smoothstep-blended) and **3D Perlin caves** in a C++/OpenGL engine (team of 3)
- **Made that world legible** — the **13-shader** GLSL pipeline I owned: 7×7 PCF shadow mapping, screen-space reflections, vertex AO, a day-night cycle and post-process water through a framebuffer pass

Mini Maya | *C++, OpenGL, Qt* — *Feb 2026 – Mar 2026*

- **Made arbitrary meshes editable in place** — a **half-edge structure** with smart-pointer ownership giving topology-aware traversal, plus an OBJ parser accepting arbitrary n-gons, rendered via OpenGL VBOs
- **Turned coarse geometry into smooth surfaces** with **Catmull–Clark subdivision** and topology ops in a Qt editor

Capsule | *React, Three.js (R3F), Blender* — *Jan 2025 – Apr 2025*

- **Made memories something you walk through rather than scroll** — the **3D gallery** in React Three Fiber for a collaborative time-capsule app (team of 4), with direct-manipulation placement of media
- **Owned how the product felt to use** across the create → contribute → unlock flow, over a Node/MongoDB API

Technical Skills

Languages: TypeScript, JavaScript, Python, Java, Kotlin, C++, C#, SQL, HTML/CSS
Frameworks & Libraries: React, Next.js, Node.js, React Native (Expo), Three.js, React Three Fiber, Jest
Infrastructure & Tools: Git, Docker, GitHub Actions, GCP, AWS, PostgreSQL/Supabase, MongoDB, PostHog, Sentry
AI & Graphics: Claude Code, Codex, Claude Agent SDK, MCP, OpenGL/GLSL, Unreal Engine 5, Unity, Maya
Practices: Trunk-based development, feature flags & staged rollout, CI/CD, code review, unit & regression testing